

Bathymetry from Space

User Guide

Prepared for AD Ports Group / Noatum Maritime - Marine Services

<https://bathymetryvmarch-production.up.railway.app/>

10 July 2026 - build aae990e

Satellite-derived bathymetry (SDB) platform for shallow coastal waters. Coordinates WGS84; vertical datum LAT (adjustable on upload); depths capped at 25 m (the physical limit of optical SDB in Gulf waters); maximum area per run 40 km².

1. Typical workflow (5 steps)

1. Draw a Region of Interest (ROI) - use the rectangle tool on the map, click a preset region button, or edit the rectangle corners. The N/S/E/W boxes in the sidebar update live.
2. Upload reference data (optional) - survey soundings (.xyz, .csv, .shp) to calibrate the model to your area. Skip this step in a predefined region: calibration is built in.
3. Choose a depth method - the resolution selector (20 m / 10 m / High-Res) for the standard run, the High-Resolution Depth panel for ~1-2 m products, the Multi-Scene MLE for a whole-year composite with uncertainty, or Wave-Based Depth where reference data is unavailable.
4. Run the extraction - click the run button. Progress appears in the LOGS tab; long high-resolution jobs appear in the JOBS window with a real percentage.
5. Validate and export - upload independent survey points to get RMSE / bias / IHO S-44 pass rates, then download the grid as GeoTIFF, CSV or GeoJSON.

2. Region of Interest

- Draw: use the rectangle draw control in the top-right of the map.
- Presets: Khalifa Port, Old Mussafah, Abu Al Abyad, Jbel Dhanna. Clicking a preset sets the ROI and the calibrated imagery date for that site, so you land on the validated optimum.
- Zoom to ROI re-centres the map on the current rectangle.
- ROIs larger than the 40 km² cap are rejected server-side; split large areas into tiles.

3. Reference / calibration data

- Formats: .xyz, .txt, .csv point lists, or Esri Shapefile (.shp + .dbf/.shx/.prj, or a .zip of them).
- Options at upload: coordinate order (Lat,Lon,Z or Lon,Lat,Z), UTM zone (EPSG) if the file is projected, horizontal datum, and vertical datum (LAT / MSL / other).
- Points falling inside the ROI are used to calibrate the depth model; a held-out fraction (20 %) is kept for honest validation.
- Predefined regions already contain survey-grade calibration - no upload needed there.

4. Run Bathymetry (standard run)

- Resolution: 20 m = fast coarse overview; 10 m = standard production depth (native Sentinel-2); High-Res = ~1 m base raster for fine detail (slower).
- The imagery used is a multi-scene cloud-free median around the chosen date, which suppresses glint, wakes and haze.
- Inside a calibrated region the run automatically uses the in-region trained model ("region-calibrated" badge); outside, a general UAE model is used with reduced weight.

5. High-Resolution Depth panel

- CLUSTERS (K): number of seabed classes used to group similar pixels before per-cluster regression. K = 8 is the calibrated default.
- Augment: automatically adds satellite-lidar and nautical-chart reference depths in bands where your calibration data is sparse.
- Imagery date: the least-cloudy scene within +/-10 days of the chosen date is selected.
- Imagery source: High-Res (~2 m/px) base raster, or Standard (10 m/px - faster, always available).
- Run High-Resolution Depth (~1 m) submits a local job (processed offline, result uploaded later). Run High-Resolution PRO (multi-scene) runs server-side and reports a real progress percentage. Both appear in the

JOBS window.

6. Multi-Scene MLE (whole year)

Combines N evenly-spaced scenes across a calendar year into a single depth product using per-pixel inverse-variance weighting (maximum likelihood). Output includes a per-pixel uncertainty map. Choose the year and 3 / 5 / 7 scenes; runtime is roughly $N \times 5\text{-}15$ s.

7. Wave-Based Depth

Reference-free bathymetry from surface-wave physics (linear dispersion): wave celerity measured from inter-band time offsets is inverted to depth. Use it where no calibration data exists and where a visible swell/wave field is present; it is not suited to flat, sheltered water.

8. Lidar depth points (ICESat-2)

Search and process satellite laser-altimeter depth points over any ROI. Results are drawn on the map colour-coded per beam and are used automatically as extra reference data by the extraction.

9. Results

- Report card (sidebar): quality badge, depth map, per-depth-band accuracy, references used, and the download row.
- Views (header): MAP (depth overlay + isobath chart tab), 3D (WebGL seabed), SPLIT (map + 3D), ASCII (text-grid preview), MONITOR, UAE RESULTS.
- RESULTS catalogue (header button): persistent list of every finished job - reopen, analyse or download past runs.

10. Validation

Upload an independent survey file (.shp/.csv) in the Validation panel, then click Run Validation. Reported: RMSE, MAE, bias, R2, IHO S-44 pass %, and the number of matched pairs. Validation points are never used for calibration.

11. Export

Every result can be downloaded as:

Format	Contents
GeoTIFF	Georeferenced depth raster (WGS84), ready for GIS/CAD
CSV	lon, lat, depth_m point list
GeoJSON	Depth points/contours for web mapping

Multi-file products (PRO runs) offer a one-click .zip of all GeoTIFFs with a manifest.

12. Monitoring

The MONITOR view stores successive surveys per zone and computes change between epochs: depth/volume differences, shoaling or deepening trends, and alerts (e.g. sudden shoaling in a navigation channel). Use it to schedule re-surveys and track dredging needs.

13. Ordering a High-Resolution product

The HIGH-RES header button opens the order panel: select an area and request a ~1 m depth product; delivery lands in the RESULTS catalogue.

Troubleshooting

- Run button greyed out - no ROI yet: draw a rectangle or click a preset region.
- "ROI too large" - the 40 km² cap; split the area.
- Noisy/blank patches - clouds or turbidity on the chosen date; move the imagery date or use the multi-scene MLE, which averages them out.
- Depths look offset by a constant - check the vertical datum chosen at upload (LAT vs MSL).
- Turbidity warning on the report - optical depth retrieval is degraded; treat results as indicative and prefer a clearer date.

Glossary

- SDB - Satellite-Derived Bathymetry.
- ROI - Region of Interest (the rectangle you draw).
- LAT - Lowest Astronomical Tide, the chart datum used for depths.
- IHO S-44 - international accuracy standard for hydrographic surveys; the pass % states how many validation points fall within the allowed error for their depth.
- MLE - Maximum-Likelihood Estimate: multi-scene, uncertainty-weighted depth composite.